

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

- A. Decreased prolactin, decreased FSH, decreased LH
- B. Decreased prolactin, increased FSH, increased LH
- C. Increased prolactin, decreased FSH, increased LH
- D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>

Finding reasoning paths:

- 1. The patient has missed her last 3 periods and has irregular menstrual bleeding .
- 2. She has spontaneous milky-white nipple discharge , which is a key symptom of hyperprolactinemia .
- 3. The symptoms are consistent with hyperprolactinemia , which can lead to amenorrhea and galactorrhea .
- 4. Hyperprolactinemia is often associated with increased prolactin levels and decreased levels of FSH and LH .

Reasoning Process:

- 1. **Clinical Presentation Analysis**:
 - The patient has missed three consecutive periods , which is indicative of amenorrhea .
 - The presence of spontaneous milky-white nipple discharge (galactorrhea) is a hallmark symptom of hyperprolactinemia .
 - Both amenorrhea and galactorrhea are common presentations of hyperprolactinemia , a condition often linked to elevated prolactin levels .
- 2. **Pathophysiology of Hyperprolactinemia**:
 - Hyperprolactinemia can result from various causes , including pituitary tumors (most common), medications , or hypothyroidism .
 - The pituitary gland regulates the release of prolactin , which inhibits the secretion of FSH and LH .
 - When prolactin levels are elevated , the negative feedback mechanism on the hypothalamus is inhibited , leading to reduced release of FSH and LH .
- 3. **Expected Laboratory Findings**:
 - Given the clinical presentation , the most likely underlying cause is hyperprolactinemia .
 - Lab results would typically show :
 - Increased prolactin levels
 - Decreased FSH (due to prolactin-induced inhibition of FSH secretion)
 - Decreased LH (due to prolactin-induced inhibition of LH secretion)

Conclusion:

The expected laboratory findings for this patient would show increased prolactin levels , decreased FSH , and decreased LH , consistent with hyperprolactinemia . This conclusion aligns with the clinical presentation and pathophysiology of hyperprolactinemia .

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<answer>

Increased prolactin , decreased FSH , decreased LH

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